

AISSAOUI Ismail

State Engineer in Artificial Intelligence & Data Science

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PROFESSIONAL SUMMARY

State Engineer graduated from ESI-SBA, specializing in **Digital Twins**, 3D visualization, and real-time AI. Expert in "Edge-to-User" latency optimization and distributed architectures for Cyber-Physical Systems (CPS). PhD candidate focused on semantic data compression for industrial Extended Reality (XR).

EDUCATION

National School of Computer Science (ESI-SBA, Algeria) 2021 – 2026
State Engineer's Degree & MSc in Artificial Intelligence

WORK EXPERIENCE

AI Engineer Intern — 3D Digital Twin Edge Computing Jan 2026 – Present
Sonatrach (Pipeline Transport Branch)

- Designed an interactive 3D monitoring dashboard (**Unity/Three.js context**).
- Optimized inference latency to **0.04ms** using **C++ JIT Compiled xLSTM**.
- Developed generative surrogates (**Mamba-SSM**) for real-time industrial simulators.
- Hardware-Software Co-design for resource-constrained mobile devices.

Network Engineering Intern Sep 2024
Algérie Télécom RMC Center

RESEARCH ENGINEERING PROJECTS

Geo-RAG — Geospatial Knowledge & Reasoning 2025
Intelligent system using RAG for semantic multi-source data processing. *Key skill: Semantic analysis and knowledge extraction.*

Recognizini — Real-Time Face Recognition & CV 2025
Active learning system with incremental embeddings for high-speed computer vision.

Career Connect — GNN recruitment Intelligence 2024 – 2025
Modeling complex relationships using Graph Neural Networks (GNN).

University Scheduling (NP-Complete Optimization) 2025
Complex system optimization using hybrid metaheuristics (**PSO/ACO/GA**).

Aura — Reinforcement Learning & Distributed Systems 2025
Distributed platform with real-time dashboards and autonomous agents.

TECHNICAL SKILLS

- XR & Graphics:** Unity 3D, Three.js, Electron, 3D Data Visualization.
- Real-Time AI:** Mamba-SSM, xLSTM, Edge Inference (C++), Latency Optimization.
- Data Science:** Semantic Learning, Data Compression, GNN, PINN, RL.
- Stack:** Python (PyTorch), C++, Docker, Node.js, Git, Linux.

LANGUAGES

Arabic (Native), English (Advanced), French (Fluent).